# Sistemas Digitais

*- Folha de Consulta para Exames -*

## Álgebra de Boole

- **Postulados**

| | | |
|---|---|---|
| **P1** | **X = 0 ou X = 1** | |
| **P2** | **0 . 0 = 0** | **1 . 1 = 1** |
| **P3** | **0 . 1 = 1 . 0 = 0** | **0 + 1 = 1 + 0 = 1** |
| **P4** | **1 + 1 = 1** | **0 + 0 = 0** |
| **P5** | $\bar{0} = 1$ | $\bar{1} = 0$ |

- **Teoremas**

| | | | |
|---|---|---|---|
| **T1** | A.0=0 | A+1=1 | |
| **T2** | A.1=A | A+0=A | |
| **T3** | A.A=A | A+A=A | **Idempotência** |
| **T4** | $A.\bar{A} = 0$ | $A + \bar{A} = 1$ | |
| **T5** | $\bar{\bar{A}} = A$ | | |
| **T6** | A.B=B.A | A+B=B+A | **Comutatividade** |
| **T7** | A.B.C=A.(B.C)=(A.B).C | A+B+C=A+(B+C)=(A+B)+C | **Associatividade** |
| **T8** | A.(B+C)=A.B+A.C | A+B.C=(A+B).(A+C) | **Distributividade** |
| **T9** | $\overline{A.B} = \bar{A} + \bar{B}$ | $\overline{A + B} = \bar{A}.\bar{B}$ | **De Morgan** |
| **T10** | A+A.B=A | A.(A+B)=A | **Absorção** |
| **T11** | $A + \bar{A}.B = A + B$ | $A.(\bar{A} + B) = A.B$ | **Termo/Factor Menor** |
| **T12** | $A.B + A.\bar{B} = A$ | $(A + B).(A + \bar{B}) = A$ | **Adjacência Lógica** |
| **T13** | $A.B + \bar{A}.C + B.C = A.B + \bar{A}.C$ | $(A + B).(\bar{A} + C).(B + C) = (A + B).(\bar{A} + C)$ | **Termo/Factor Incluído** |

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## Flip-Flops - Tabelas funcionais e de excitação

- **Flip-Flop SR**

| S | R | Qn+1 |
|---|---|---|
| 0 | 0 | Qn |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | "Indefinido" |

| Qn → Qn+1 | | S | R |
|---|---|---|---|
| 0 | 0 | 0 | X |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | X | 0 |

- **Flip-Flop D**

| D | Qn | Qn+1 |
|---|---|---|
| 0 | X | 0 |
| 1 | X | 1 |

| Qn → Qn+1 | | D |
|---|---|---|
| X | 1 | 1 |
| X | 0 | 0 |

- **Flip-Flop JK**

| J | K | Qn+1 |
|---|---|---|
| 0 | 0 | Qn |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | $\overline{Qn}$ |

| Qn → Qn+1 | | J | K |
|---|---|---|---|
| 0 | 0 | 0 | X |
| 0 | 1 | 1 | X |
| 1 | 0 | X | 1 |
| 1 | 1 | X | 0 |

- **Flip-Flop T**

| T | Qn+1 |
|---|---|
| 0 | Qn |
| 1 | $\overline{Qn}$ |

| Qn → Qn+1 | | T |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |